An induction principle for the integers with the three computation rules, derived from the idempotent retract of N x N

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:sign-induction' (source lines 1399-1414); the proof environment begins on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's remark Z-quotient-by-canonical-representatives (source lines 1380-1397), which defines the idempotent r used in the first line of the proof; without it the phrase 'the above idempotent' at source line 1416 has no referent inside the package.

Editorial note. The retained counter values (chapter 6, section 10, theorem offset 10) make the retained remark print as Remark 6.10.11 and the counted lemma as Lemma 6.10.12, matching the original numbering; remark, lemma and corollary share one counter in the book's macro file.

Editorial note. No retained line contains a citation, an included file or a non-ASCII character, so no bibliography and no asset is declared.

Editorial scope. One original result of the HoTT Book, counted once: An induction principle for the integers with the three computation rules, derived from the idempotent retract of N x N.

Allowed standard prerequisites (editorial).

- The integers as a set quotient of N x N. Z is presented as the set quotient of N x N by the relation (a,b) ~ (a',b') iff a + b' = a' + b, and for a set A with an idempotent r : A -> A whose fixed points satisfy the canonical-representative condition the book identifies A/~ with the type of fixed points.. The proof replaces Z by the retract Sm{x:N x N}(r(x) = x) at source line 1416 and pulls the given data back along the quotient map q (source lines 1418-1425).
- List-free double induction on N x N and the diagonal case. A family Q : N x N -> Type can be inhabited by recursion on the two arguments separately, the diagonal successor case being the one that is not automatic.. The function g is built by double induction with the four clauses g(0,0), g(suc n,0), g(0,suc m) and g(suc n,suc m) (source lines 1428-1434).
- Transport of data across equalities, and equivalences induced by equal elements of a quotient. Given d : P(z) and a path z = z', transport gives

an element of P(z'); and if q(x) = q(x') in the quotient, the induced map between the fibres is an equivalence.. The transport step at source lines 1420-1425 produces d'_0, d'_+ and d'_-, and the equivalence e_{n,m} at source lines 1426-1427 is induced by q(n,m) = q(suc n, suc m).

Source reference. Original lemma 6.10.8: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1554–1554. Pointer to the original lemma that identifies a set with the fixed points of an idempotent; the retained proof invokes it to replace Z by the retract and then carries out the double induction itself.

Source reference. Original section 8: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 167–167. Pointer to the chapter where the resulting induction principle is applied later.

Remark 6.10.11. original lemma 6.10.8 applies to $\mathbb{Z}$ with the idempotent $r : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ defined by

$$r(a,b) = \begin{cases} (a-b,0) & \text{if } a \geq b, \\ (0,b-a) & \text{otherwise.} \end{cases}$$

(This is a valid definition even constructively, since the relation $\geq$ on $\mathbb{N}$ is decidable.) Thus a non-negative integer is canonically represented as $(k,0)$ and a non-positive one by $(0,m)$, for $k,m : \mathbb{N}$. This division into cases implies the following "induction principle" for integers, which will be useful in original section 8. (As usual, we identify a natural number n with the corresponding non-negative integer, i.e. with the image of $(n,0) : \mathbb{N} \times \mathbb{N}$ in $\mathbb{Z}$.)

Lemma 6.10.12. *Suppose $P : \mathbb{Z} \to \mathcal{U}$ is a type family and that we have*

- $d_0 : P(0)$,
- $d_+ : \prod_{(n:\mathbb{N})} P(n) \to P(\mathsf{succ}(n))$, *and*
- $d_- : \prod_{(n:\mathbb{N})} P(-n) \to P(-\mathsf{succ}(n))$.

Then we have $f : \prod_{(z:\mathbb{Z})} P(z)$ such that

- $f(0) = d_0$,
- $f(\mathsf{succ}(n)) = d_+(n, f(n))$ *for all* $n : \mathbb{N}$, *and*
- $f(-\mathsf{succ}(n)) = d_-(n, f(-n))$ *for all* $n : \mathbb{N}$.

Proof. For purposes of this proof, let $\mathbb{Z}$ denote $\sum_{(x:\mathbb{N}\times\mathbb{N})}(r(x) = x)$, where r is the above idempotent. (We can then transport the result to any equivalent definition of $\mathbb{Z}$.) Let $q : \mathbb{N} \times \mathbb{N} \to \mathbb{Z}$ be the quotient map, defined by $q(x) = (r(x), i(x))$ as in original lemma 6.10.8. Now define $Q :\equiv P \circ q : \mathbb{N} \times \mathbb{N} \to \mathcal{U}$. By transporting the given data across appropriate equalities, we obtain

$$\begin{aligned} d'_0 &: Q(0,0) \\ d'_+ &: \prod_{n:\mathbb{N}} Q(n,0) \to Q(\mathsf{succ}(n),0) \\ d'_- &: \prod_{n:\mathbb{N}} Q(0,n) \to Q(0,\mathsf{succ}(n)). \end{aligned}$$

Note also that since $q(n,m) = q(\mathsf{succ}(n), \mathsf{succ}(m))$, we have an induced equivalence

$$e_{n,m} : Q(n,m) \simeq Q(\mathsf{succ}(n), \mathsf{succ}(m)).$$

We can then construct $g : \prod_{(x:\mathbb{N}\times\mathbb{N})} Q(x)$ by double induction on x:

$$\begin{aligned} g(0,0) &:\equiv d'_0, \\ g(\mathsf{succ}(n), 0) &:\equiv d'_+(n, g(n,0)), \\ g(0, \mathsf{succ}(m)) &:\equiv d'_-(m, g(0,m)), \\ g(\mathsf{succ}(n), \mathsf{succ}(m)) &:\equiv e_{n,m}(g(n,m)). \end{aligned}$$

Now we have $\mathsf{pr}_1 : \mathbb{Z} \to \mathbb{N} \times \mathbb{N}$, with the property that $q \circ \mathsf{pr}_1 = \mathsf{id}$. In particular, therefore, we have $Q \circ \mathsf{pr}_1 = P$, and hence a family of equivalences $s : \prod_{(z:\mathbb{Z})} Q(\mathsf{pr}_1(z)) \simeq P(z)$. Thus, we can define $f(z) = s(z, g(\mathsf{pr}_1(z)))$ to obtain $f : \prod_{(z:\mathbb{Z})} P(z)$, and verify the desired equalities. □